CPU

Datei: CPU.kicad_sch

Video-Generation

Datei: Video-Generation.kicad_sch

Color-Modulator

Datei: Color-Modulator.kicad_sch

RF-Stage VHF

Datei: RF-Stage VHF.kicad_sch

RF-Stage UHF

Datei: RF-Stage UHF.kicad_sch

| Symbol | Note |
|---|---|
| 7Vrms 1 | Measurement point with measured value |
| 1 | Adjustment point |
| 1 | Curve shape (oscillogram) |
| 1 | |
| 1 | Component or Circuit Note |
| | Module/Cassette |
| | PCB |
| | Functional group |

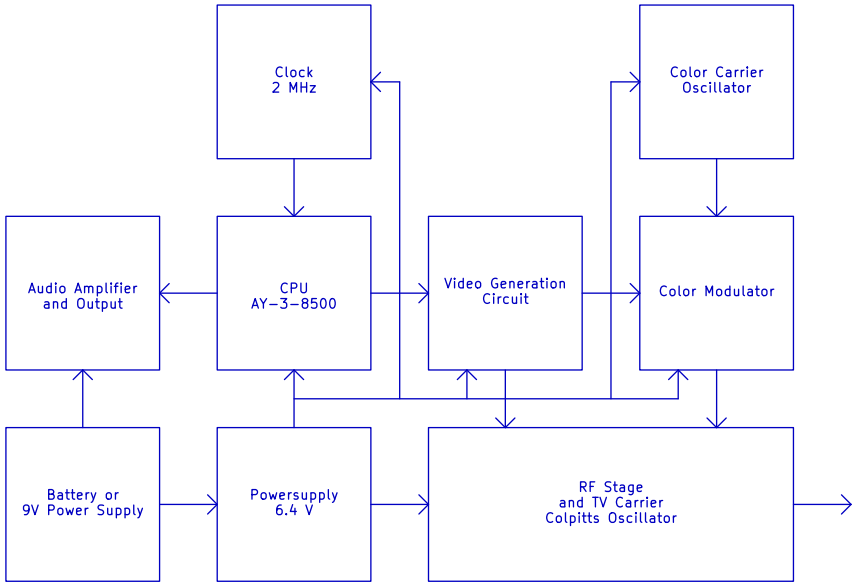
Note regarding component names:

All component designations written in red were either missing, illegible, or duplicated. I attempted to deduce their meaning logically and have noted below which designations this applies to.

- C1 assigned because it is unreadable
- C4 assigned because it is unreadable
- C9 assigned because there is no label
- C42 assigned because it is unreadable
- C101 assigned because it is unreadable
- C106 assigned because it is unreadable
- D21 assigned because it is duplicated (D20)
- FL1, FL2, FL3 assigned because they are unreadable
- Jx assigned because there is no label
- L101 assigned because it is unreadable
- LS1 assigned because there is no label
- Q101 assigned because it is unreadable
- RV1, RV2 assigned because there are no labels
- U1, U2, U3 assigned because they are unreadable
- XTAL1 assigned because it is unreadable

All measurements with the oscilloscope were taken with a 1:10 probe to avoid loading the RF circuits.

LINK TO REPOSITORY



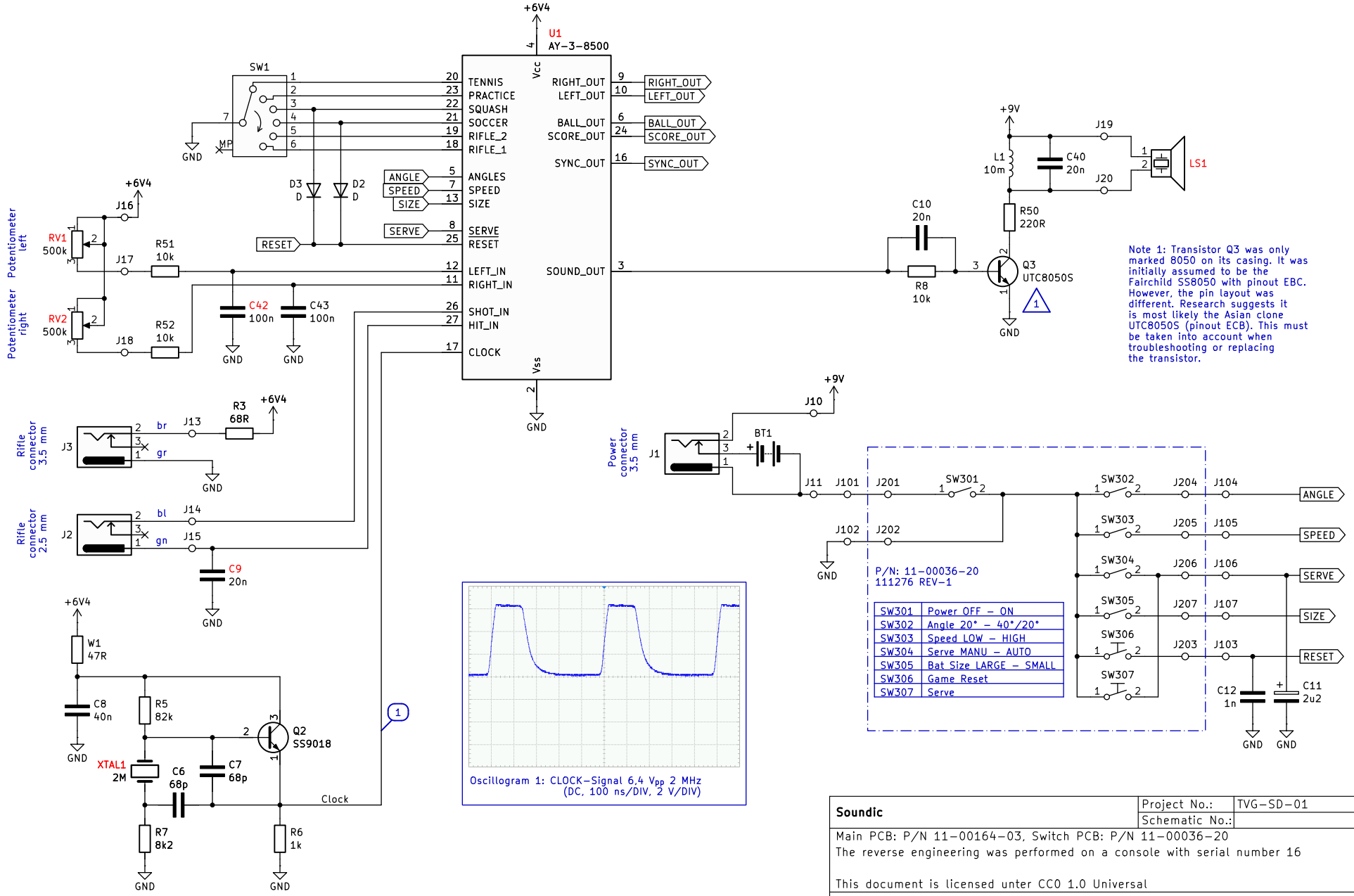

| | | | |
|---|---|---|---|
| **Soundic** | | Project No.: | TVG-SD-01 |
| | | Schematic No.: | |

Main PCB: P/N 11-00164-03, Switch PCB: P/N 11-00036-20
The reverse engineering was performed on a console with serial number 16

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***Title: TVS SD-01 PAL Color Console (Pong Clone)***

| | | | | | |
|---|---|---|---|---|---|
| Size: | A4 | KiCad E.D.A. 10.0.8 | Author: M. Fiedler | Date: | 2026-10-30 |
| Sheet: | / | | | Rev: | 1.0 |
| File: | kicad_schematic.kicad_sch | | | Page: | 1/6 |



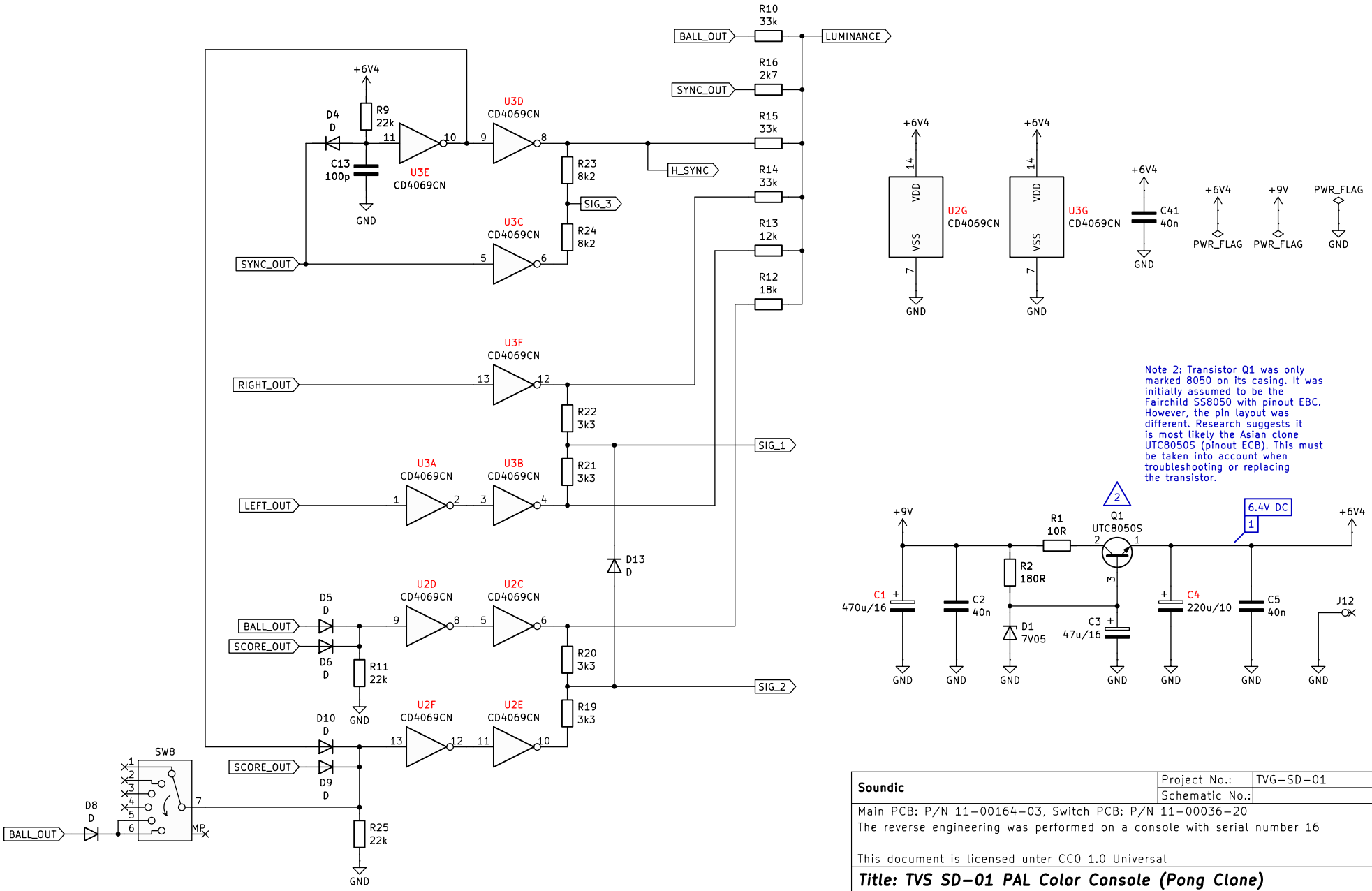

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| File: | CPU.kicad_sch | | | Page: | 2/6 |




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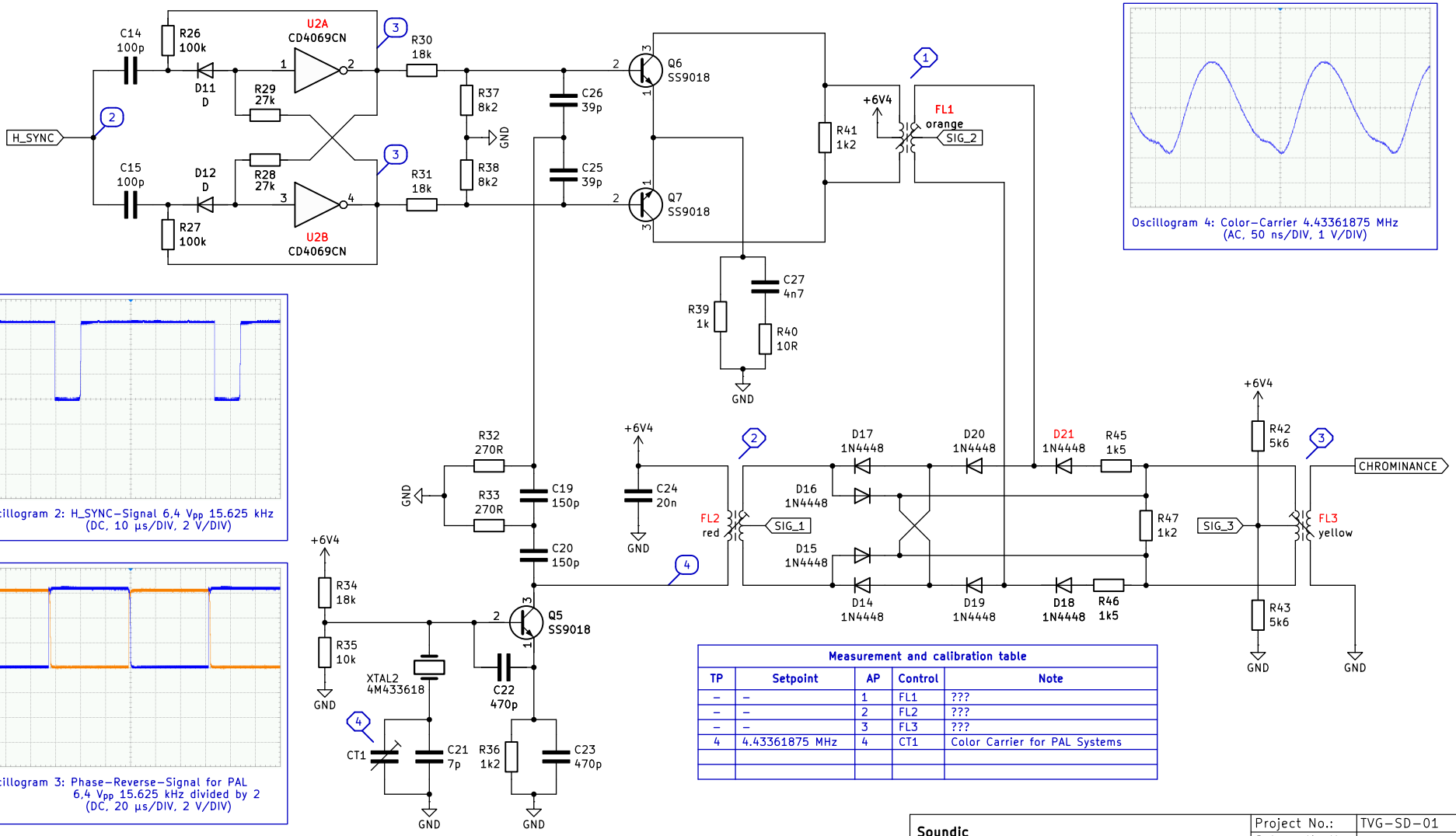
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| Size: | A4 | KiCad E.D.A. 10.0.8 | Author: M. Fiedler | Date: | 2026-10-30 |
| Sheet: | /Video-Generation/ | | | Rev: | 1.0 |
| File: | Video-Generation.kicad_sch | | | Page: | 3/6 |

T–Flip–Flop
Frequency Divider






Oscillogram 2: H_SYNC–Signal 6,4 $V_{pp}$ 15.625 kHz
(DC, 10 µs/DIV, 2 V/DIV)



Oscillogram 3: Phase–Reverse–Signal for PAL
6,4 $V_{pp}$ 15.625 kHz divided by 2
(DC, 20 µs/DIV, 2 V/DIV)



Oscillogram 4: Color–Carrier 4.43361875 MHz
(AC, 50 ns/DIV, 1 V/DIV)

**Measurement and calibration table**

| TP | Setpoint | AP | Control | Note |
|---|---|---|---|---|
| – | – | 1 | FL1 | ??? |
| – | – | 2 | FL2 | ??? |
| – | – | 3 | FL3 | ??? |
| 4 | 4.43361875 MHz | 4 | CT1 | Color Carrier for PAL Systems |
| | | | | |
| | | | | |

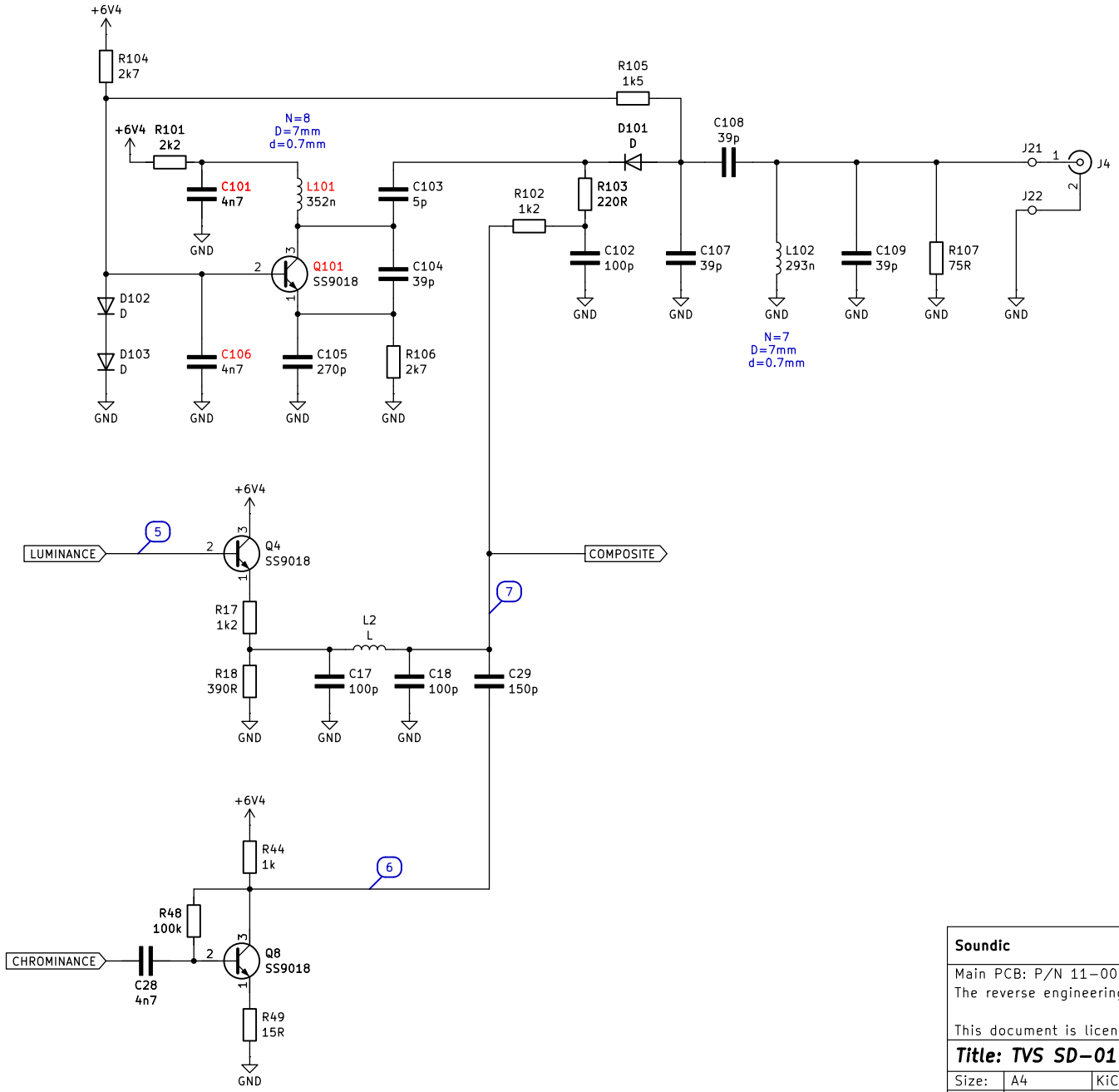
| | | | |
|---|---|---|---|
| **Soundic** | Project No.: | TVG–SD–01 | |
| | Schematic No.: | | |

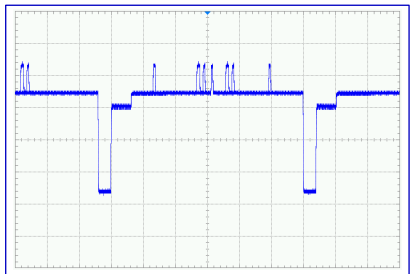
Main PCB: P/N 11–00164–03, Switch PCB: P/N 11–00036–20
The reverse engineering was performed on a console with serial number 16

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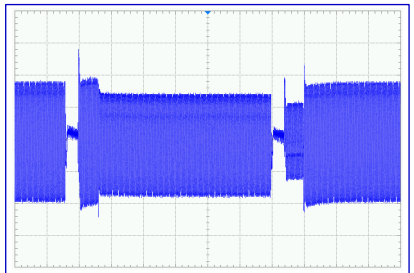
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| Size: | A4 | KiCad E.D.A. 10.0.8 | Author: M. Fiedler | Date: | 2026–10–30 |
| Sheet: | /Color–Modulator/ | | | Rev: | 1.0 |
| File: | Color–Modulator.kicad_sch | | | Page: | 4/6 |

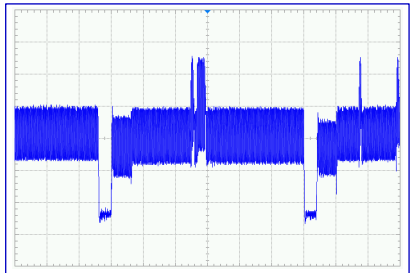


Oscillogram 5: LUMINANCE–Signal (Video–Signal)
(DC, 10 µs/DIV, 1 V/DIV)



Oscillogram 6: CHROMINANCE–Signal
(AC, 10 µs/DIV, 200 mV/DIV)



Oscillogram 7: LUMINANCE + CHROMINANCE–Signal
(DC, 10 µs/DIV, 200 mV/DIV)

| | | | |
|---|---|---|---|
| **Soundic** | | Project No.: | TVG–SD–01 |
| | | Schematic No.: | |

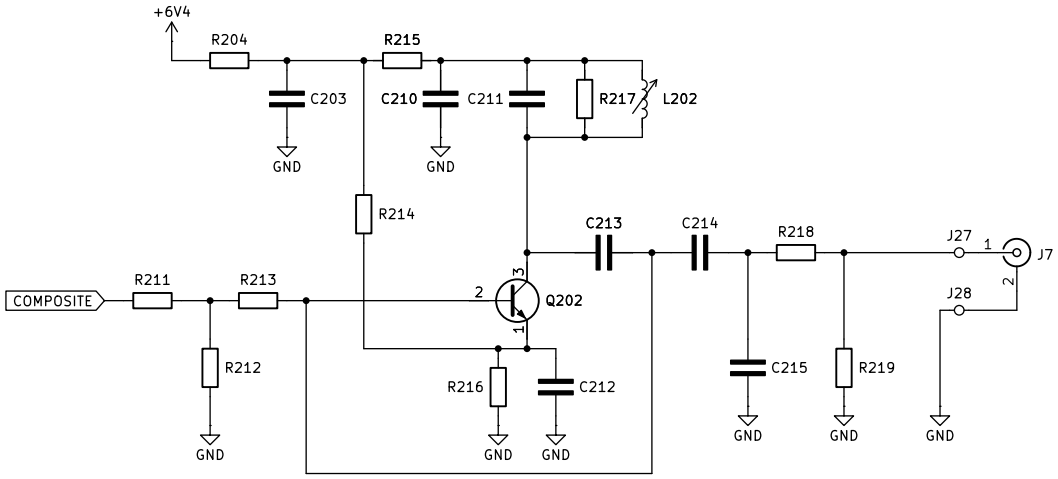
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| Sheet: | /RF–Stage VHF/ | | | Rev: | 1.0 |
| File: | RF–Stage VHF.kicad_sch | | | Page: | 5/6 |

The UHF stage was created based on the component labels on the circuit board. This section is unpopulated in my console, so I cannot provide component values or guarantee the accuracy of the circuit.




| | | | |
|---|---|---|---|
| **Soundic** | | Project No.: | TVG-SD-01 |
| | | Schematic No.: | |

Main PCB: P/N 11-00164-03, Switch PCB: P/N 11-00036-20
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| File: | RF-Stage UHF.kicad_sch | | | Page: | 6/6 |